

Console Tester ROM modification guide

This single guide covers every current expansion of the SHIFT838/Geoff Trott Console Tester V1: the working 16K diode adapter, permanent 16K LS21 version, two clean 32K choices, homebrew 32K diode option, practical 40K maximum that retains tester RAM, and the more difficult 48K and banked alternatives. Because the tester is used by itself, the mappings prioritize dependable diagnostic ROM space rather than coexistence with a cartridge, DSR, or ordinary 32K expansion.

Read this first

The PCB artwork and photographs establish the intended routes; continuity on the assembled board remains the final authority. Switch the console off before inserting or removing the sidecar or EPROM, and make resistance/continuity measurements only with the board disconnected.

Never directly join 74LS138 outputs: they are active-low push-pull outputs. Combine them only with the LS21 or documented Schottky-diode network.

The current adapter has a 28-pin W27C512 with four pins lifted or folded clear of the U3 socket:

W27C512 pin	Signal	Connection
1	A15	U2 pin 3, TI A0, address weight >8000
20	/CE	New combined ROM-select node
26	A13	U2 pin 1, TI A2, address weight >2000
27	A14	U2 pin 2, TI A1, address weight >4000

All other pins remain in the socket: pin 28 is +5 V, pin 14 is ground, and pin 22 /OE retains the board's read-qualified path. From above, pin 1 is upper left with the single notch at top. Prefer a turned-pin carrier to repeatedly bending the W27C512's own legs.

Board and decoder map

The photographed 2024 PCB contains:

Reference	Fitted device	Relevant function
U1	SN74LS174AN	Six-bit indicator latch
U2	74LS138N	Eight active-low 8K address selects
U3	28-pin 27C64 socket	Original 8K diagnostic ROM
U4	74LS373N	Board latch/buffer
U5	74ALS02N	NOR logic
U6	74ALS00N	NAND logic
U7	AS6C6264-55PCN	8K tester SRAM at >C000->DFFF

There is no LS21 on the photographed PCB; any LS21 in this guide is an added part. U2 decodes TI A0-A2 as follows:

U2 output	U2 pin	CPU range	Normal use in this project
Y0	15	>0000->1FFF	Console CPU ROM; never claim
Y1	14	>2000->3FFF	Optional ROM window
Y2	13	>4000->5FFF	Optional ROM window; normally DSR space
Y3	12	>6000->7FFF	16K diagnostic extension
Y4	11	>8000->9FFF	Scratchpad/MMIO; never claim
Y5	10	>A000->BFFF	Optional ROM window
Y6	9	>C000->DFFF	Existing U7 tester SRAM
Y7	7	>E000->FFFF	Fixed boot core and LOAD vector

The W27C512 uses conventional address numbering while TI names the most significant CPU address bit A0. The direct linear mapping is therefore:

```
W27 A0  <- TI A15      byte weight >0001
...
W27 A12 <- TI A3       byte weight >1000
W27 A13 <- TI A2       byte weight >2000
W27 A14 <- TI A1       byte weight >4000
W27 A15 <- TI A0       byte weight >8000
```

Changing only A13-A15 does not add a CPU window. It makes W27 file offsets match CPU addresses. The `/CE` decoder determines which 8K CPU windows are actually visible.

Options at a glance

Configuration	Direct ROM	Added parts	Selected windows	Recommendation
Original/repeated W27	8K	none beyond carrier wiring	Y7	Electrical bring-up only
W27 two-diode trial	16K	2 Schottky diodes, 2.2K	Y3, Y7	Proven temporary setup
W27 + LS21	16K	1 SN74LS21, bypass capacitor	Y3, Y7	Permanent 16K version
W27 + LS21	32K	same LS21, one gate	Y1, Y3, Y5, Y7	Clean W27 expansion
W27 four-diode trial	32K	4 Schottky diodes, 2.2K	Y1, Y3, Y5, Y7	Homebrew/qualification only
27C256 + LS21	32K	1 SN74LS21, bypass capacitor	Y1, Y3, Y5, Y7	Simplest dedicated 32K board
W27 + cascaded LS21 gates	40K	same one LS21 package	Y1, Y2, Y3, Y5, Y7	Maximum clean direct ROM with U7
W27 + Y6	48K	decoder still fits one LS21	Y1, Y2, Y3, Y5, Y6, Y7	Experimental; requires U7/workspace redesign

The 16K, 32K, and 40K configurations retain U7 SRAM and the diagnostic's fail-safe workspace. The 40K configuration is still relatively clean: one W27C512, one LS21 package, and additional decoder wires. The 48K configuration is where the design becomes substantially more complicated because ROM and U7 would otherwise drive `>C000->DFFF` at the same time.

Scenario 1: current temporary 16K W27C512 diode adapter

This is the physically proven public-beta configuration: Y3 and Y7 feed a pulled-up `/CE` node through two Schottky diodes, exposing the extension at `>6000` and the fixed core at `>E000` while retaining U7 SRAM at `>C000`.

The complete illustrated parts list, pin-by-pin wiring sequence, diode orientation, continuity checklist, programming steps, and first-power test now have one authoritative home in the [v0.8 operator and modification manual](#). Use that procedure for the working two-diode build. The remaining scenarios in this document are alternatives for permanent or larger-capacity boards.

Scenario 2: permanent 16K W27C512 with one SN74LS21

The SN74LS21 contains two four-input positive-AND gates. Because U2 selections are active low, ANDing Y3 and Y7 produces a low ROM `/CE` in either selected range:

```
/CE_ROM = Y3 AND Y7 AND HIGH AND HIGH
```

Wire the first gate:

LS21 pin	Connection
1	U2 pin 12/Y3, CPU <code>>6000->7FFF</code>
2	U2 pin 7/Y7, CPU <code>>E000->FFFF</code>
4, 5	Defined HIGH, directly to +5 V or through a shared 1K-4.7K pull-up
6	Lifted W27 pin 20 <code>/CE</code>
7	Ground
14	+5 V
9, 10, 12, 13	Ground unused second-gate inputs
8	Leave unused second-gate output open
3, 11	Package no-connect pins; leave open

Fit a 0.1 uF ceramic capacitor directly between LS21 pins 14 and 7. The three W27 high-address wires remain exactly as shown at the start of this guide.

Scenario 3: 32K W27C512 with one LS21 gate

This is the clean way to keep the W27C512 and expose 32K. Use four decoder outputs in the first LS21 gate:

```

LS21 pin 1 <- U2 pin 14/Y1 >2000
LS21 pin 2 <- U2 pin 12/Y3 >6000
LS21 pin 4 <- U2 pin 10/Y5 >A000
LS21 pin 5 <- U2 pin 7/Y7 >E000
LS21 pin 6 -> lifted W27 pin 20 /CE

```

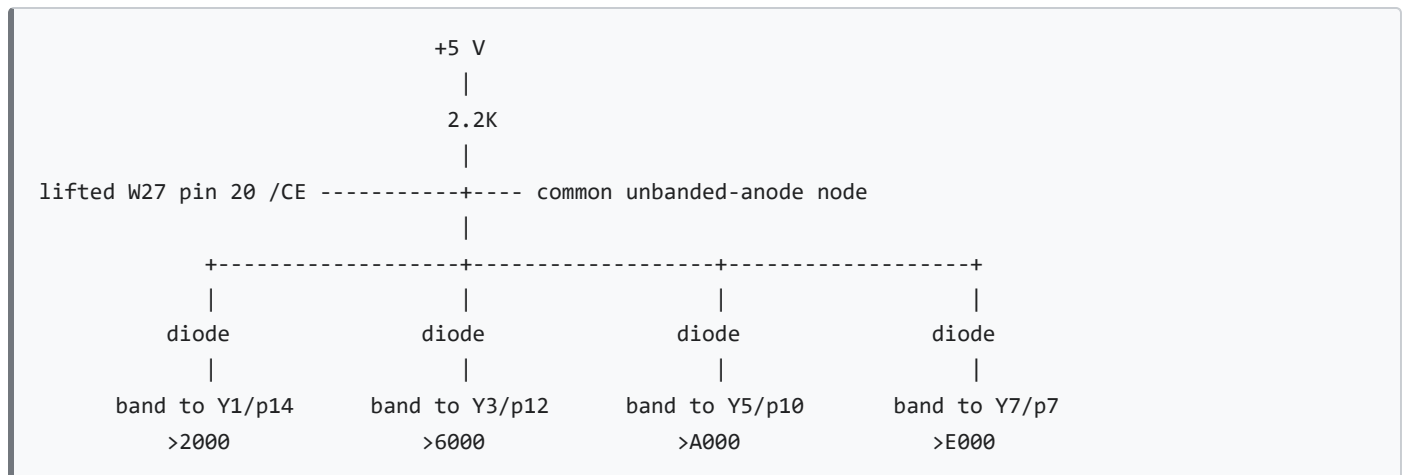
The W27 address wires stay unchanged, so the sparse 64K programmer file uses the same offsets as CPU addresses:

W27 file offsets	CPU range
>2000->3FFF	>2000->3FFF
>6000->7FFF	>6000->7FFF
>A000->BFFF	>A000->BFFF
>E000->FFFF	>E000->FFFF

U7 remains at >c000 , and DSR space at >4000 remains unclaimed. The tester must still be used without a normal 32K expansion or cartridge because those devices contend with ranges already owned by the tester.

Homebrew four-diode 32K variant

A temporary 32K W27 adapter can use four matched Schottky diodes instead of the first LS21 gate. Keep the same common pulled-up /CE anode node used by the working 16K circuit and add banded cathode branches to U2 Y1, Y3, Y5, and Y7:



Four branches add leakage and capacitance. Re-measure /CE HIGH and LOW in every selected and unselected window and inspect edges with a scope. This is a valid homebrew experiment, not the recommended production circuit.

Scenario 4: dedicated 32K 27C256 with one LS21

If 32K is the final target, a JEDEC-pinout 27C256 is the simplest dedicated solution. It needs only three isolated EPROM pins:

27C256 pin	Signal	Connection
20	/CE	LS21 pin 6
26	A13	U2 pin 2, TI A1
27	A14	U2 pin 3, TI A0

Pin 1 remains seated only after U3 socket pin 1 is verified as +5 V, because the 27C256 requires VPP at VCC during reads. All other pins remain in the socket.

Use the first LS21 gate:

LS21 pin	Connection
1	U2 pin 14/Y1, >2000
2	U2 pin 12/Y3, >6000
4	U2 pin 10/Y5, >A000
5	U2 pin 7/Y7, >E000
6	Lifted 27C256 pin 20 /CE
7, 14	Ground, +5 V respectively

As with the 16K circuit, bypass pins 14 and 7 with 0.1 uF and tie every unused TTL input to a defined level.

The 27C256 programmer file is a packed 32K image, not a sparse W27 image:

27C256 file offsets	Visible at CPU
>0000->1FFF	>2000->3FFF
>2000->3FFF	>6000->7FFF
>4000->5FFF	>A000->BFFF
>6000->7FFF	>E000->FFFF

Never program a 64K W27C512 image unchanged into this layout. Generate and verify a separately packed 32768-byte image.

Scenario 5: 40K W27C512 with the same LS21 package

Forty kilobytes is the maximum directly visible ROM that retains U7 tester SRAM. It occupies >2000, >4000, >6000, >A000, and >E000. Five active-low inputs require both four-input gates inside the one LS21 package:

```

gate 1:
  LS21 pin 1 <- U2 pin 14/Y1  >2000
  LS21 pin 2 <- U2 pin 13/Y2  >4000
  LS21 pin 4 <- U2 pin 12/Y3  >6000
  LS21 pin 5 <- U2 pin 10/Y5  >A000
  LS21 pin 6 -> X

gate 2:
  LS21 pin 9  <- X from pin 6
  LS21 pin 10 <- U2 pin 7/Y7   >E000
  LS21 pins 12 and 13 <- defined HIGH
  LS21 pin 8 -> lifted W27 pin 20 /CE

```

The W27 high-address wiring and same-offset 64K file layout remain unchanged. This uses DSR space at `>4000`, which is acceptable for a standalone tester. Validate the additional gate delay and all five `/CE` cases on hardware.

Scenario 6: why direct 48K is different

Adding U2 Y6 would expose a sixth block at `>C000->DFFF`, but U7 already owns that range. Selecting both devices would cause bus contention. A 48K version must disable or remove U7 and rewrite the firmware so its LOAD workspace, early RAM test, and fail-safe LED reporting no longer depend on `>C000`.

The decoder still fits in the existing LS21 cascade by replacing one of the second gate's HIGH inputs with Y6, but that fact does not make the complete 48K conversion simple. Do not wire Y6 with the current firmware.

Banked alternatives

Banking leaves the CPU map mostly intact and pages several physical 8K W27 blocks through one aperture. It needs a reset-defined latch and bank-aware software, so it is not easier than the current direct 16K arrangement.

Stored ROM	Bank bits	Typical added storage logic
16K	1	One 74LS74 section or one wider-latch bit
32K	2	Two defined latch bits
64K	3	74LS378, 74LS174, or equivalent three-bit latch

Execute the bank-change trampoline from U7 RAM, provide a known entry in every bank, and retain a valid boot/LOAD vector in every possible power-up bank unless the latch has a proven reset state. A bank latch must not interfere with the existing LED-latch write at `>E002`.

Programmer images and staged bring-up

For the current 16K W27 hardware, use these project outputs:

File	Purpose
<code>build/ti99-sidecar-diag-w27c512.bin</code>	First electrical test: eight identical copies of the 8K core
<code>build/ti99-sidecar-diag-beta-0.8-16k-w27c512.bin</code>	Current sparse 16K build: extension at <code>>6000</code> , fixed core at <code>>E000</code>

File	Purpose
build/HEXDIAG08.BIN	Byte-identical TI-safe short filename for the Beta 0.8 programmer image

Select W27C512 in the programmer, erase and blank-check it, program the entire 65536-byte file, and run verify. Do not treat either raw 8K component image as an already positioned 64K programmer file. Record the SHA-256 from `build/manifest.json` for the exact build placed in a device.

Bring up any new circuit in stages:

1. Reconfirm the unmodified board with its known-working 27C64.
2. Check every lifted pin and wire with power off.
3. Use the repeated-core image first. Confirm boot, LOAD, LED walk, screen, and U7 RAM before executing from a new window.
4. Confirm `/OE` is inactive on writes, especially the LED write at `>E002`.
5. Probe `/CE` in each intended and unintended 8K range.
6. Program the correctly laid-out final image only after the electrical test passes.
7. For the current 16K build, `16K EXTENSION PRESENT` proves the A13-A15 mapping, Y3/Y7 selection, extension signature, and full extension checksum. `CORRUPT` indicates a real programming, decoding, or ROM read-path fault.

Do not attach a cartridge, memory expansion, or another sidecar during initial qualification.

Unpowered verification checklist

For a W27C512 carrier, verify:

- W27 pins 1, 20, 26, and 27 do not contact their same-numbered U3 pads;
- W27 pin 1 reaches only U2 pin 3 among the three address pickups;
- W27 pin 27 reaches U2 pin 2;
- W27 pin 26 reaches U2 pin 1;
- W27 pin 20 reaches only the new `/CE` output/common node;
- W27 pin 22 remains connected to the original `/OE` path;
- W27 pins 28 and 14 reach +5 V and ground respectively;
- selected U2 outputs are not shorted to one another;
- no lifted contact can touch its former pad or an adjacent pin; and
- resistance between +5 V and ground does not indicate a short.

For an LS21 circuit, also verify pin 14 to +5 V, pin 7 to ground, the correct gate output to lifted EPROM pin 20, defined levels on every unused input, and the 0.1 uF bypass capacitor. For a diode circuit, verify diode polarity and the 2.2K path from the common `/CE` node to +5 V.

Capacity decision

- Keep the working **16K diode** adapter for immediate firmware development.
- Use the **16K LS21** circuit for a permanent version of the proven map.
- Choose **32K W27C512** when retaining the existing device and direct file offsets matters.

- Choose **32K 27C256** for the smallest clean dedicated fixed-capacity board.
- Choose **40K W27C512** when one logic package and a few more wires are worth the extra space; it is the practical maximum without sacrificing U7.
- Treat **48K direct** and **banked ROM** as later redesigns, not incremental changes to the current firmware.

See the [technical reference](#) for the board's base memory use and firmware contract independent of the physical modification.